Panel IMS: Information and Management System for Market Research

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1. INTRODUCTION

1.1. Background of the System/Organization

Businesses play a crucial role in driving economic development. To remain competitive, organizations must engage in continuous decision-making and problem-solving as part of their daily operations. However, navigating the market has become increasingly challenging due to rapid technological advancements and persistent uncertainties in the external environment. The COVID-19 pandemic serves as a notable example of an unprecedented disruption that forced many businesses to close and accelerated significant, lasting changes in consumer behavior and market dynamics. In recent years, the rapid shift from traditional brick-and-mortar stores to online marketplaces has transformed the competitive landscape, particularly in countries with dynamic and evolving markets such as the Philippines. While this transition has provided consumers with greater convenience and flexibility in their shopping experiences, it has also intensified competition among businesses. Consequently, firms increasingly rely on market research and data-driven insights to better understand market conditions, reduce uncertainty, and make informed strategic decisions that support long-term success.

Market research empowers businesses to make informed choices, stay competitive, and better satisfy customer needs. It supports businesses by gathering, analyzing, and interpreting information on target customers (e.g., demographics, needs, preferences, behavior), competitors or players, trends, and the products or services of a particular market. Such information helps a business better understand the market environment so that it can ultimately make informed, strategic decisions on product development, marketing strategies, pricing, customer experience, and potential business expansion.

Market-research companies collect such information in two ways: **primary research**, which directly collects data from sources such as surveys, interviews, focus groups, panel data, and point-of-sale tracking; and **secondary research**, which uses existing data such as reports, statistics, and industry publications. Market-research companies that employ primary research and focus on panel and point-of-sale (POS) data follow structured methodologies to keep their panel current and to ensure their samples stay representative, high-quality, and unbiased — one of them through panel (or sample) update.

Panel update is a regular process in which company vendors conduct manual store visits to re-evaluate shops and retailers already in their panel, while also profiling new shops that have not yet been accounted for. This sample is then extrapolated given the company's market knowledge and what they estimate the actual population to be.

The system developed in this project stemmed from the need of a multinational market-research company to streamline this sample-update process, which is done at least once a year. Its local office in the Philippines focuses on point-of-sale (POS) and panel data for retail measurement services (RMS). The company uses proprietary data-warehousing software and uses ETL (extract-transform-load) for a large bulk of the tasks carried out by its local Data Science team. The idea of an information system that acts as the source of truth for shop profiles and extrapolation information may well be integrated with the ETL already in place to streamline the overall panel-update pipeline. The app aims to facilitate panel refresh, improve the accuracy of market projections, and ensure seamless integration of new data for ongoing market research. This maintains the credibility of the panel-based insights that the company sells to its vendors and brand clients.

1.2. Problem Statement

The panel-update process at an RMS (Retail Measurement Service) market-research company involves managing vast amounts of shop information and point-of-sale (POS) data across multiple markets and product groups. In many traditional companies, this process is handled manually through spreadsheets and fragmented extrapolation setups. As the number of shops and market segments fluctuates, maintaining accuracy becomes increasingly difficult. The complexity of simulation, coordination between teams, and manual data handling often leads to inconsistencies, errors, and operational delays. This project addresses these challenges by proposing the development of a centralized, deployable system to streamline panel-data management and minimize human error.

1.3. Project Objectives

The main objective of this project was to develop a robust information and management system app that is easily configurable and shareable among members of the team for collaborative tasks. Specifically, the study aimed to:

- digitize and centralize panel update and management across markets to reduce manual intervention, data errors, and inconsistencies;
- integrate automated market simulation of the extrapolation setup into the same system;
- enable a map-based visualization of stores in the panel for inspection and analysis; and
- provide real-time monitoring of panel-health metrics and simulated market trends through a comprehensive dashboard.

The app is expected to be used readily by the subject market-research company to help it manage and maintain its large database of shops and retailers.

1.4. Project Scope, Limitations, and Assumptions

This project focused on the development of an information and management system with four main features: manage shop databases, retrieve extrapolation factors, locate stores on a map, and view key panel-volume metrics in a dashboard. The backend script handles the extrapolation logic as shops are added, deleted, or modified, using a simple logic fed to the backend model. Extrapolation factors are updated whenever there are changes — whether from adding new shops or updating the profile information of existing ones. This is because extrapolation factors are calculated based on a predefined extrapolation cell, which segments data for more accurate projections. Therefore, any change to the sample count and segment will result in changes to the extrapolation factors. The app allows users to view current extrapolation factors and download them as needed.

This paper excluded the more complex analytics and extrapolation-optimization steps of the standard sample-update pipeline, where extrapolations are fine-tuned to minimize the share of missing retail shops — below the Standard General Protocol (SGP) threshold — without causing significant impact on current brand share and trends.

2. REVIEW OF RELATED LITERATURE

There is little information about the actual solutions that other market-research companies employ in their panel update, but solutions to the same problem depend on, and are usually tailored to, the sampling methodology used. This section discusses three third-party software products available in the market that market-research companies could invest in for panel management. The use of these solution tools allows companies to digitize their data collection and panel study, which shifts the gear away from traditional sampling designs.

Qualtrics is a software product that market-research companies could invest in for panel management. It is used widely in academia for survey research and has since pivoted toward serving enterprise customers. Qualtrics offers a comprehensive panel-management system that automates the recruitment, targeting, and engagement of panel members [1].

Nebu Dub InterViewer is a data-collection solution that provides users with face-to-face, telephone, and internet market research using a single platform. It enables users to deliver surveys to mobile devices via browser or through its Nebu native apps. It has a flexible, intuitive interface that accelerates online survey scripting and creates engaging questionnaires. It provides users with Nebu CAPI, a mobile app that distributes surveys and features GPS positioning, audio, video, and photo capture. Users can then stream these datasets directly to its data hub, where they can collect, manage, utilize, and enrich data with predictive analytic calculations [2].

ARCS by Marketing Systems Group is another panel-management platform that is more panel-member-centric. It optimizes panel engagement by providing panel members with features to self-schedule and respond to studies. It leverages the cloud to deliver projects quickly, improve panel management, and better secure panel data [3].

3. SYSTEMS ANALYSIS AND DESIGN

3.1. Requirements Analysis

This project delivers a readily functional application developed in Python using the Dash library. The underlying datasets were created, hosted, and generated in pgAdmin via PostgreSQL.

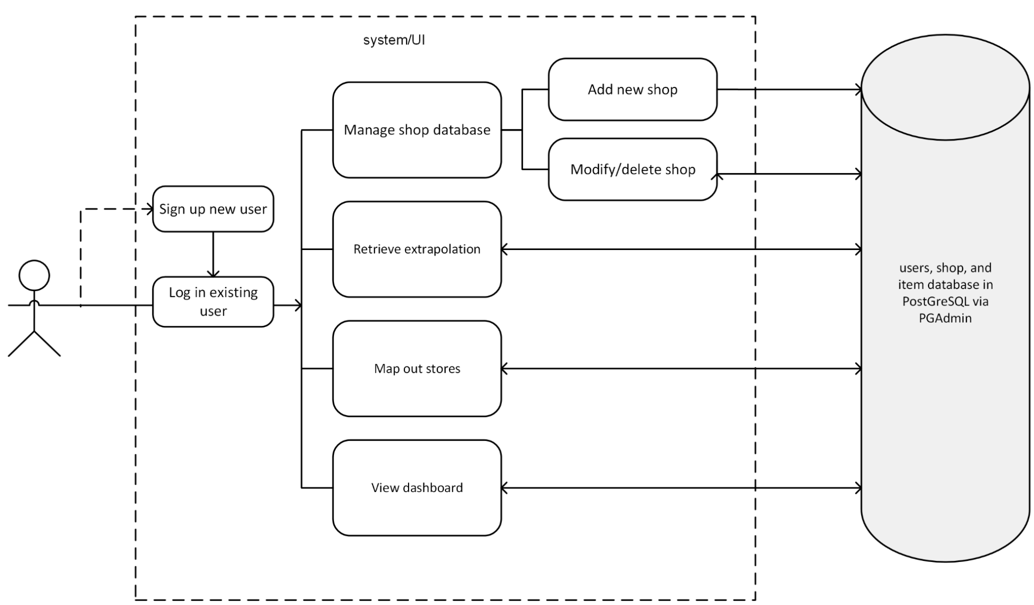

Fig 1. *Systems Design of Panel IMS*

3.1.1. Functional Analysis

3.1.1.1 User authentication

Upon execution, the app prompts users either to log in using their credentials (registered username and password) if they have an existing account, or to create an account as a new user via the sign-up page.
Once logged in, the homepage shows four main modules: (1) manage shop data, (2) retrieve extrapolation information, (3) display stores on a map, and (4) view the prebuilt dashboard.

3.1.1.2. Shop data management

The main tasks of auditors during a regular panel update are to add new shops, modify the profile of shops whose category and segment have changed since the last market audit, and tag closed shops as inactive in the shop database. This ensures that samples in the panel are up to date and tagged in the correct segment based on the latest audit. The shop-data manager has two submodules — *Add New Shop* and *Modify/Delete Shop* — which together support three actions:

1. **Add New Shops and Profile Information:** the app allows users to input information about newly identified shops that were previously unaccounted for.
2. **Delete Inactive or Closed Shops:** the app can soft-delete shops that have been flagged as inactive or closed and have no presence in the market.
3. **Edit Existing Shop Profiles:** the app enables users to update the profile information of shops already in the database, keeping the data current.

Adding a shop requires a user-defined Outlet ID that has not been used before. It also asks for the store address's geocode (longitude and latitude), which is a crucial reference for future store visits. The add module also lets users insert the products and sales that the shop carries. The system uses this historical sales data to simulate market trends and brand share using the current extrapolation setup, which can be fine-tuned based on market knowledge.

3.1.1.3. Extrapolation retrieval

One major challenge in the market-research industry is to estimate the market population as accurately as possible. Collected samples attempt to generalize the population by way of extrapolation. The company uses a specific logic to update extrapolation factors in the two applicable scenarios: addition of new shops, and segment transfer of a shop due to a profile change. The factor is calculated per segment — called an extrapolation cell — and is identical for every shop within that cell. This segment (or cell) ideally varies by product group and market. For Singapore's dishwashers product group, the company employs a shopsize–organization-type–channel–region segment for its extrapolation setup.

Figure (a): Addition of new shop D — assumes shops A, B, C, D are all the shops in an extrapolation cell

Cell Current	
Shop	Factor
A	3
B	3
C	3

Cell New	
Shop	Factor
A	2.5
B	2.5
C	2.5
D	2.5

$$EF_{new} = \frac{3+3+3+1}{4} = 2.5$$

Figure (b): Segment change of shop D

Cell 1 Current	
Shop	Factor
A	2.5
B	2.5
C	2.5
D	2.5

Cell 2 Current	
Shop	Factor
E	1
F	1
G	1

Cell 1 New	
Shop	Factor
A	2.5
B	2.5
C	2.5

Cell 2 New	
Shop	Factor
E	1.375
F	1.375
G	1.375
D	1.375

$$EF_{new}(cell\ 2) = \frac{1+1+1+2.5}{4} = 1.375$$

Since this project uses point-of-sale (POS) data for the simulation, extrapolation inspection and fine-tuning are crucial, as they affect the accuracy of the projected sales data of brand clients in relation to their competitors and the overall market. The full list of outlets and their extrapolation factors can be viewed and downloaded as CSV through the download button in this module.

3.1.1.4. Store mapper

Incorporating store-location mapping into the panel study provides valuable insights into the representativeness, variance, and potential sparsity of collected samples, enabling more strategic planning for audits and future data collection. This app features a map-based visualization of stores already in the panel, which can be viewed and filtered by segment and attribute.

3.1.1.5. Dashboard viewer

The application also offers a preconfigured two-tab dashboard designed to support auditors and managers in routinely tracking key volume metrics related to their panel and the market simulation based on its current extrapolation setup. The Market Simulation tab provides quick insights into the extrapolated sales units and value (in SGD), which are typically validated using the company's knowledge of the market about brand rankings and top-performing products. This is particularly crucial for brand clients who are critical about their market position and ranking against competitors. The results can be incorporated into a deck for high-level discussion with the company's senior leadership.

3.1.2. Non-Functional Analysis

The Sign-Up page should throw an error if the account being registered already exists with the same employee ID or username. The extrapolation viewer, store mapper, and dashboard should all provide a real-time view into the current panel data and extrapolation setup.

3.2. *Entity-Relationship Diagram*

The database comprises five core domain entities — shops, retailers, items, extrapolation cell, and sales — alongside a separate *registeredusers* table that supports authentication. These entities carry the following key attributes: outlet ID, item ID, item name, brand, product group (main category of item), retailer, copies (indicates whether sales are actual or modeled), shop size (size category based on assortment and floor size), channel (e.g., Electrical Specialists, Technical Superstores), and region (among the five regions in Singapore).

The extrapolation-cell entity contains the extrapolation factor for each distinct segment, calculated based on the predefined segment designed by the company. For the given product group and market, the company calculates the unique extrapolation factor by the shopsize–organization-type–channel–region segment. The sales entity lists the aggregated units and average price (in SGD) over three consecutive months of 2021 sales, where sales units were extrapolated based on the factors generated.

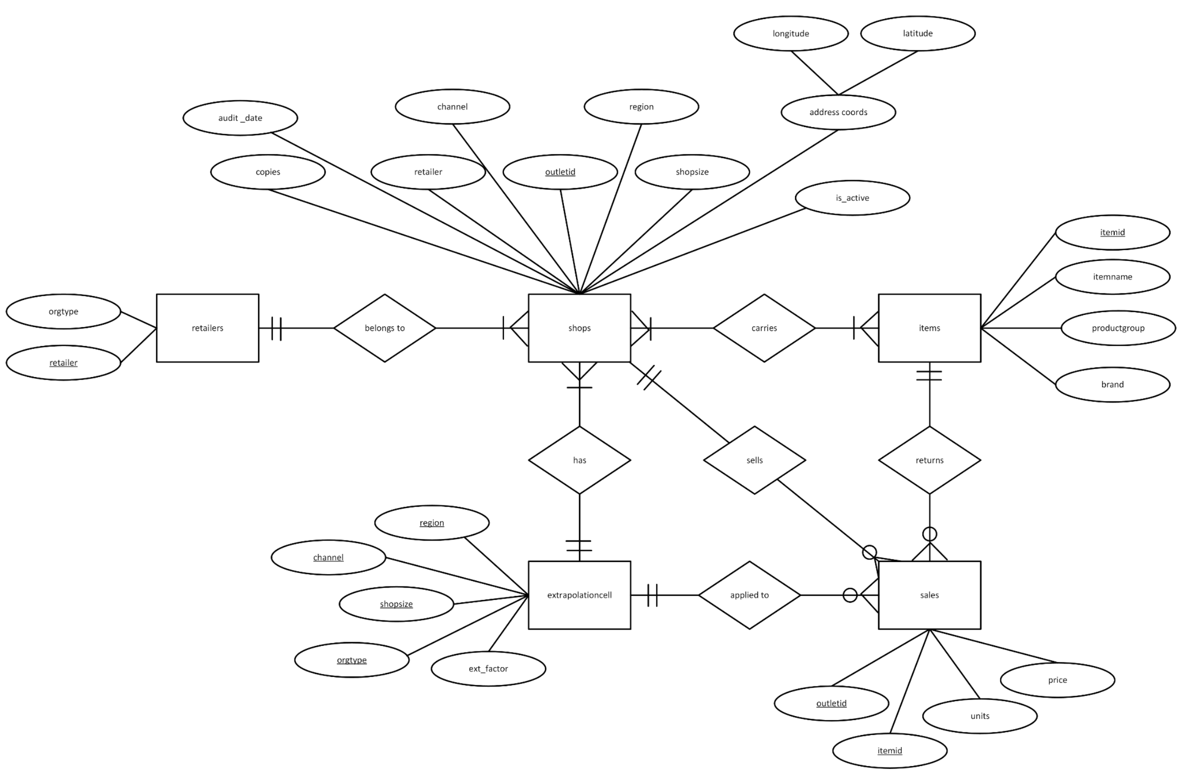

Fig 2. *ERD of Panel IMS*

3.3. Relational Database Model

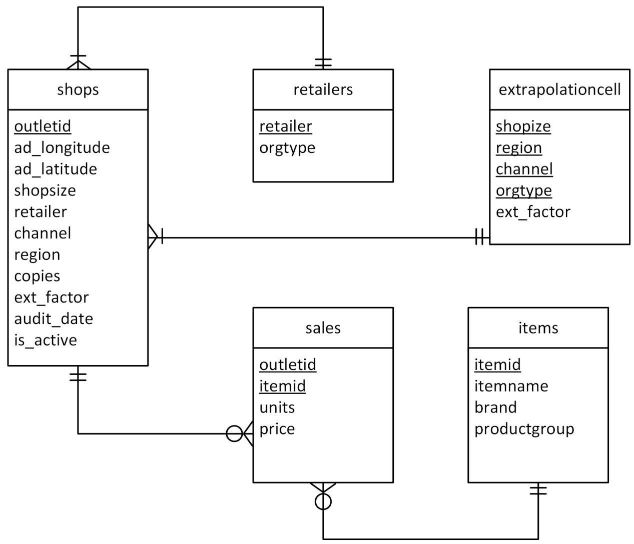

Fig 3. *RD Model of Panel IMS*

3.4 Data Sources

The dataset used in the implementation involves both panel data and point-of-sale (POS) data. The panel data was generated from a subset of total Singapore shops that registered sales for dishwashers in the months of June to August 2021. The POS data is a subset of actual three-period sales for Singapore's dishwasher product group, with all attributes retrieved at once from a market-research company's proprietary MDM data-warehouse portal in 2021. Sales data were aggregated and slightly modified to ensure anonymity and to honor the confidentiality agreement with the retailers involved.

4. PHYSICAL DATABASE DESIGN

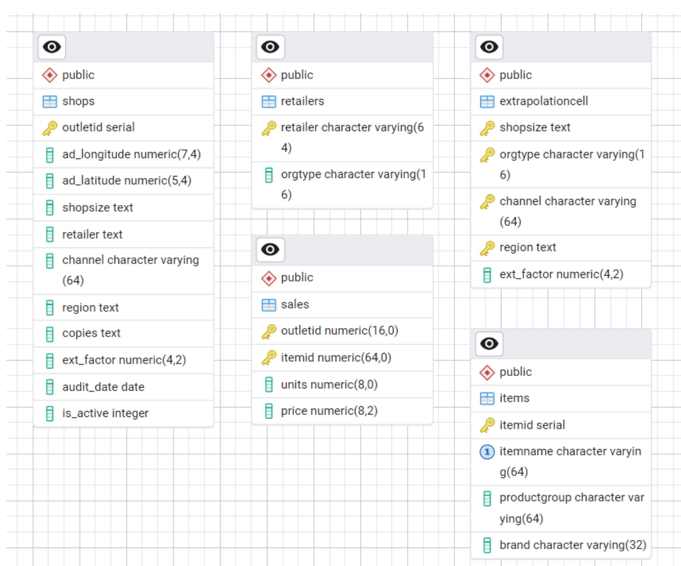

Fig 4. *ERD of Panel IMS DB generated by pgAdmin*

```
1  SELECT * FROM shops
```

Data Output   Messages   Notifications

Showing rows: 1 to 56   Page No:

	outletid [PK] integer	ad_longitude numeric (7,4)	ad_latitude numeric (5,4)	shopsize text	retailer text	channel character varying (64)	region text	copies text	ext_factor numeric (4,2)	audit_date date	is_active integer
1	1427	103.8024	1.3412	XLARGE	TANGS	Department Stores	CENTRAL REG ...	COPY_AND_CREATE	1.00	2025-01-07	1
2	1867	103.8452	1.3115	XLARGE	BHG	Department Stores	CENTRAL REG ...	COPY_AND_CREATE	1.00	2025-04-21	1
3	4884	103.8333	1.2988	LARGE	COURTS	Electrical Specialists	CENTRAL REG ...	REGULAR	1.00	2025-01-04	1
4	4601	103.8063	1.3158	LARGE	HARVEY NORMAN	Electrical Specialists	CENTRAL REG ...	COPY_AND_CREATE	1.00	2024-12-14	1
5	2795	103.8057	1.3364	LARGE	GAIN CITY	Electrical Specialists	CENTRAL REG ...	REGULAR	1.00	2025-04-21	1
6	3060	103.8031	1.3508	LARGE	BEST DENKI	Electrical Specialists	CENTRAL REG ...	REGULAR	1.00	2024-10-09	1
7	4885	103.8102	1.3451	LARGE	BEST DENKI	Electrical Specialists	CENTRAL REG ...	REGULAR	1.00	2024-10-11	1

Fig 5. *Entity 'shop' in pgAdmin*

```
1  SELECT * FROM retailers
```

Data Output   Messages   Notifications

Showing rows: 1 to 11

	retailer [PK] character varying (64)	orgtype character varying (16)
1	MUSTAFA	INDEPENDENTS
2	TANGS	INDEPENDENTS
3	BHG	INDEPENDENTS
4	COURTS	CHAINS
5	BEST DENKI	CHAINS
6	PARISILK ELECTRONICS	CHAINS
7	HARVEY NORMAN	CHAINS
8	GAIN CITY	CHAINS
9	MAYER MARKETING	INDEPENDENTS
10	AUDIO HOUSE	CHAINS
11	MEGA DISCOUNT	CHAINS

Fig 6. *Entity 'retailers' in pgAdmin*

```
1  SELECT * FROM extrapolationcell
```

Data Output   Messages   Notifications

Showing rows: 1 to 17

	shopsize [PK] text	orgtype [PK] character varying (16)	channel [PK] character varying (64)	region [PK] text	ext_factor numeric (4,2)
1	LARGE	CHAINS	Electrical Specialists	EAST REGION SG	1.00
2	XLARGE	CHAINS	Electrical Specialists	NORTH EAST SG	1.00
3	LARGE	CHAINS	Technical Superstores	NORTH EAST SG	1.00
4	XSMALL	CHAINS	Electrical Specialists	NORTH EAST SG	1.00
5	SMALL	CHAINS	Electrical Specialists	NORTH EAST SG	1.00
6	LARGE	CHAINS	Electrical Specialists	NORTH EAST SG	1.25
7	LARGE	CHAINS	Technical Superstores	CENTRAL REG ...	1.00
8	LARGE	CHAINS	Electrical Specialists	CENTRAL REG ...	1.00
9	XLARGE	CHAINS	Technical Superstores	NORTH REG SG	1.00
10	LARGE	CHAINS	Electrical Specialists	WEST REG SG	1.00
11	XLARGE	CHAINS	Technical Superstores	EAST REGION SG	1.00

Fig 7. *Entity 'extrapolationcell' in pgAdmin*

```
1  SELECT * FROM items
```

Data Output | Messages | Notifications

Showing rows: 1 to 58

	itemid [PK] integer	itemname character varying (64)	productgroup character varying (64)	brand character varying (32)
1	149976736	DFB425FP	DISHWASHERS	LG
2	158043996	EBDW 1351 A WH	DISHWASHERS	ELBA
3	146597283	DFN05X11W	DISHWASHERS	BEKO
4	171641414	WFC3C26XAUS	DISHWASHERS	WHIRLPOOL
5	156338670	EDW 3050U	DISHWASHERS	EUROPACE
6	162803615	SMS46GW01P	DISHWASHERS	BOSCH
7	97823566	SMS63L02EA	DISHWASHERS	BOSCH
8	58664210	SMS50E82EU	DISHWASHERS	BOSCH
9	167766976	SMS2HAI12E	DISHWASHERS	BOSCH

Fig 8. *Entity 'items' in pgAdmin*

```
1  SELECT * FROM sales
```

Data Output | Messages | Notifications

Showing rows: 1 to 622

	outletid [PK] numeric (16)	itemid [PK] numeric (64)	units numeric (8)	price numeric (8,2)
1	725	97823566	3	956.33
2	725	122212884	2	1295.80
3	725	124184210	1	859.00
4	725	133569980	2	759.00
5	725	134027714	1	1309.00
6	725	140199652	1	1399.00
7	725	143315315	1	1394.00
8	725	147284348	3	1739.00

Fig 9. *Entity 'sales' in pgAdmin*

5. IMPLEMENTATION AND INTERFACE DESIGN

5.1. Log In and Sign Up pages

The application requires approved user credentials to log into the portal. Each account is uniquely identified by its employee ID, with the username and email also enforced as unique values; the sign-up email must end in @mail.com. Login is authenticated using the registered username and password.

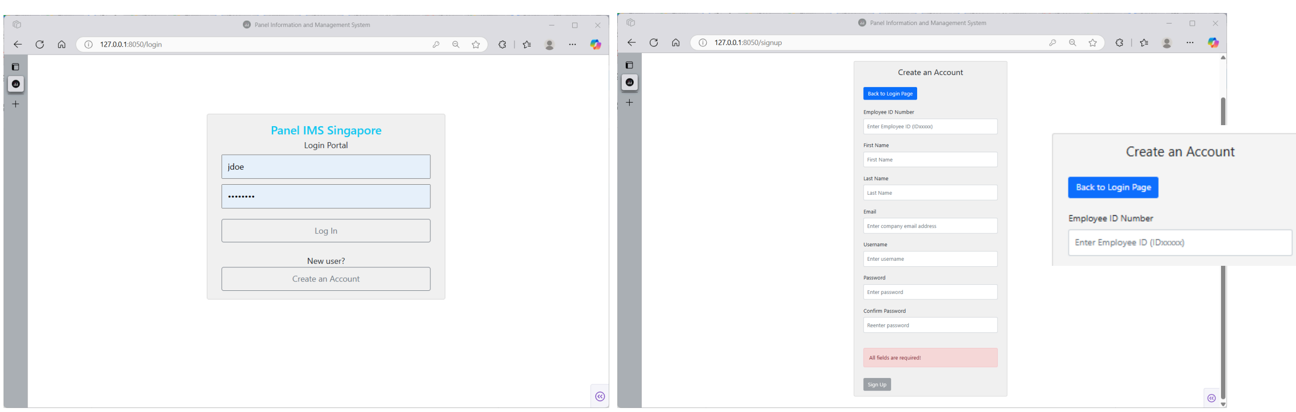

Fig 10. *Log In (left) and Sign up (right) pages*

5.2. Home page

Panel IMS's homepage displays its four modules (main features) a user can choose from. Clicking anywhere within the bounds of a module's box redirects the user to the corresponding module.

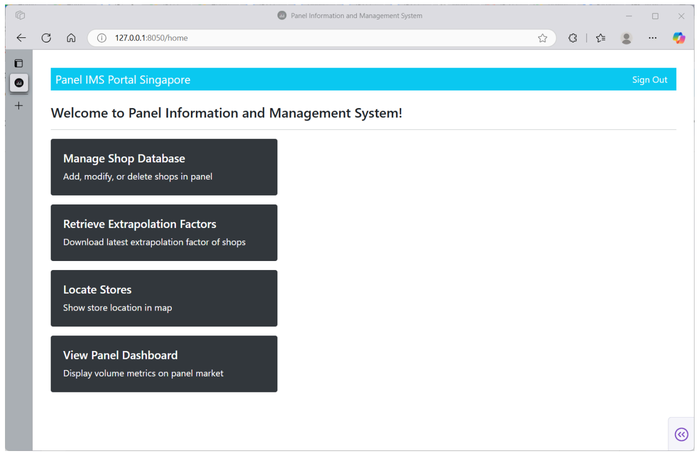

Fig 11. *Panel IMS homepage*

5.3. Module: Manage Shop Database page

This module is a powerful feature of Panel IMS that lets a user modify the underlying database hosted in pgAdmin straight from the app itself. Two functions are provided: *Add New Shop* and *Modify/Delete Shop*. Each button redirects to another page where forms are provided to create, modify, or delete database records.

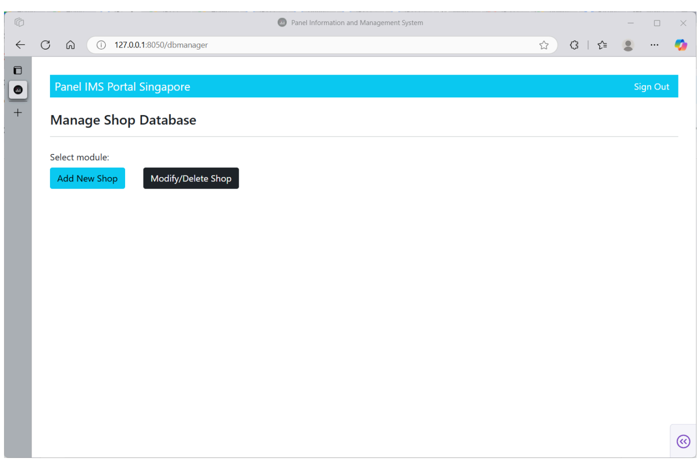

Fig 12. *Panel IMS shop manager page*

5.4. Submodule: Add New Shop page

This is the page shown when the *Add New Shop* button is clicked from the previous page. It contains two sections: Shop Profile and Products Carried.

The Shop Profile section requires the user to fill in all the fields provided. This section combines a user-defined input box, dropdown menus, a date picker, and value-constrained input fields for address coordinates. One important step in the universe (panel) update of the subject company is geocoding the collected store addresses; online geocoding tools, such as Google Maps, support multi-location geocoding.

The Products Carried section can be used to insert items and their historical sales based on store visits or audits. The tabular layout gives the user the flexibility to view or delete items that were already inserted. This section also includes input boxes for sales units and price, which default to zero if left unfilled.

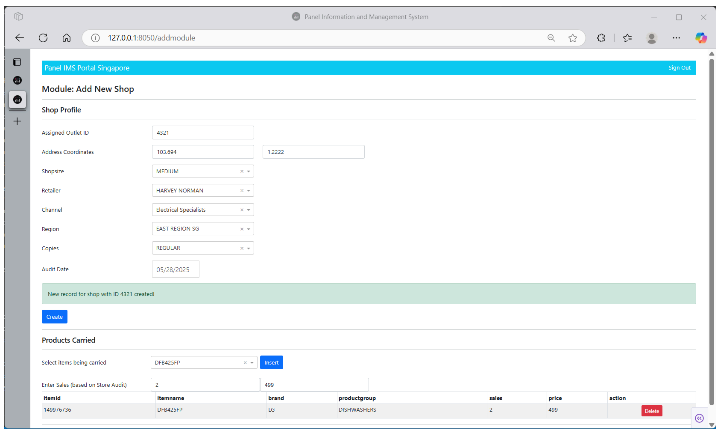

Fig 13. *Add New Shop module page of Panel IMS*

5.5. Submodule: Modify/Delete Shop pages

This submodule is used for editing the information of existing shops and deleting them as needed. It supports a tabular layout of the records for a user to inspect before making such aggressive data changes. It also has a search box that can be used to filter the data and locate the shop in question efficiently.

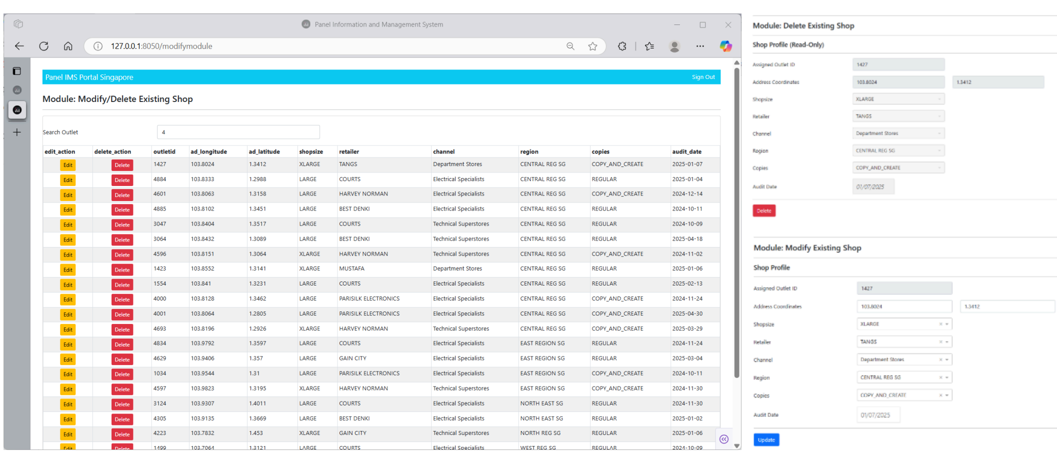

Fig 14. *Modify/Delete Shop module pages of Panel IMS*

5.6. Module: Extrapolation page

This page returns the updated and recalculated extrapolation factors of all shops in response to any new changes made to the data by addition or modification of shop records. It supports a search box for more flexibility and efficiency. It also includes a button that allows a user to download the table as a CSV file. This becomes handy for simulation exercises and data analysis performed externally.

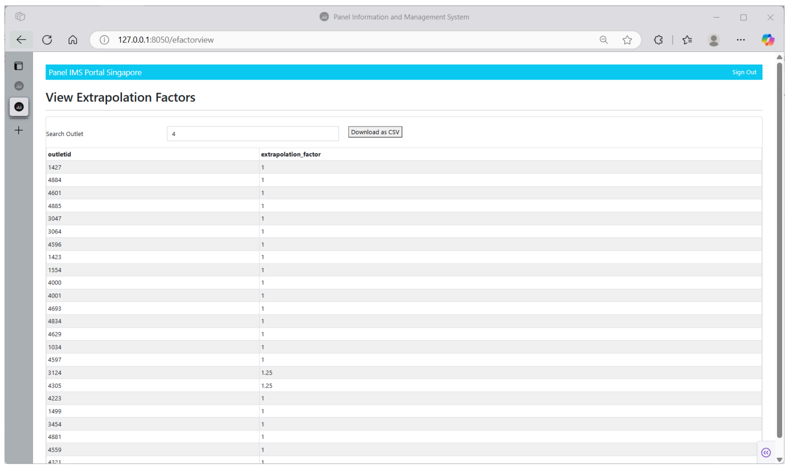

Fig 15. *Extrapolation page of Panel IMS*

5.7. Module: Store mapper page

The store mapper page is another powerful feature of Panel IMS that visually locates all or a subset of stores on a map. The filters at the top allow users to customize the search results by segment or attribute value.

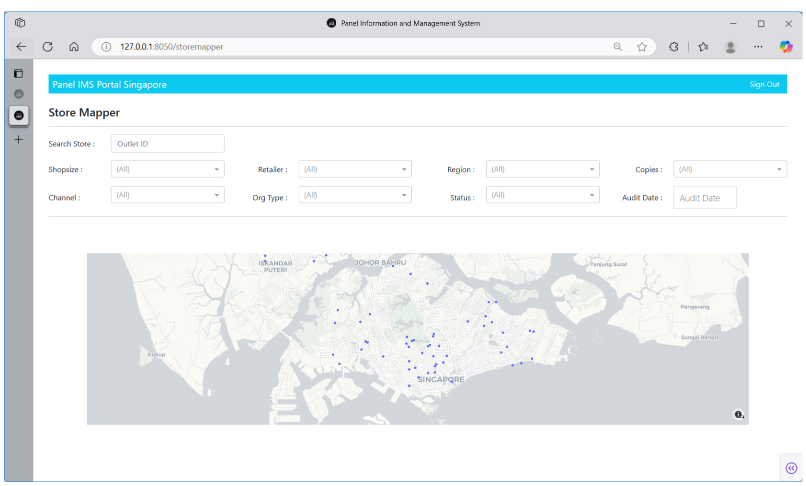

Fig 16. *Store mapper page of Panel IMS*

5.8. Module: Dashboard page

This page displays the two-tab panel dashboard with relevant key metrics for instant, real-time access and monitoring. The Market Simulation tab provides results of the simulated extrapolation setup in terms of brand- and item-level trends. The visuals can be segmented or filtered using the dropdowns and search box above the charts.

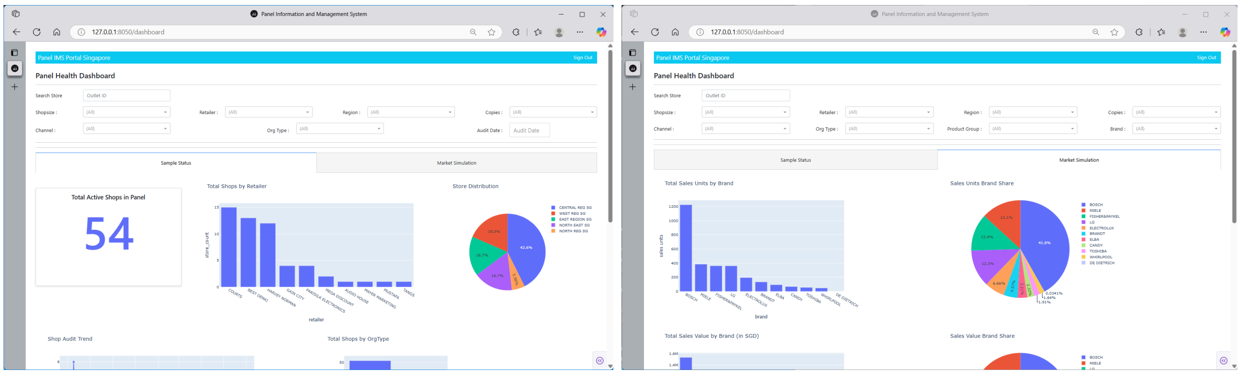

Fig 17. *Panel dashboard page of Panel IMS*

All pages of the Panel IMS app display the navigation bar, which can be clicked either to move one page back or to sign out of the portal. Clicking the "Panel IMS Portal Singapore" header moves the page one step back, while "Sign out" on the right end allows the user to end the session at any time and on any page.

6. REPORTS

Panel IMS can be seamlessly integrated into the analytical toolkits of analysts, market researchers, and managers to support informed business decision-making. In particular, users can leverage Panel IMS in the following ways.

6.1. Downloadable Extrapolation Factor table

Extrapolation factors are readily available for download and can be used for external analysis and routine reporting. Future enhancements may include the development of additional underlying tables to support more comprehensive external analyses and reporting needs. Moreover, the project could be expanded to incorporate API integration, enabling automation of the data-extraction process.

6.2 Store mapper feature

Incorporating store-location mapping from a sample or subset enhances the toolkit available to market researchers during panel studies. This feature provides valuable insights into the representativeness, variance, and potential sparsity of collected samples, enabling more strategic planning for audits and future data collection. The application includes a map-based visualization of panel stores, which can be filtered by segment and field for targeted analysis.

6.3. Panel health dashboard

The application offers a preconfigured two-tab dashboard designed to support auditors and managers in routinely tracking key volume metrics related to their panel, and to provide real-time market simulation of their current extrapolation setup. The Sample Status tab provides a summary of the total number of stores by segment and presents a line chart illustrating audit activity over time. The Market Simulation tab generates the simulation results of extrapolation in terms of total-market, brand-, and item-level impact of the current setup. These can be incorporated into a deck for discussion with senior leadership.

7. REFERENCES

[1] Qualtrics. (n.d.). *Online sample*. <https://www.qualtrics.com/en-au/research-services/online-sample/>
[2] GetApp. (n.d.). *Dub InterViewer Reviews 2024: Details, pricing, & features*. <https://www.getapp.com/customer-management-software/a/dub-interviewer/>
[3] msg systems. (n.d.). *ARCS*. <https://www.m-s-g.com/Pages/arcs/>

8. APPENDICES

8.1. PostgreSQL Code

```
CREATE TABLE IF NOT EXISTS public.registeredusers
(
	emp_id VARCHAR(7) PRIMARY KEY,
	username VARCHAR(16) UNIQUE NOT NULL,
	userpassword VARCHAR(32) NOT NULL,
	firstname VARCHAR(64) NOT NULL,
	lastname VARCHAR(64) NOT NULL,
	email VARCHAR(64) UNIQUE NOT NULL,
	created_timestamp TIMESTAMP DEFAULT NOW(),
	is_active INT DEFAULT 1,
		CHECK (is_active IN(0,1))
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.registeredusers
   OWNER to postgres;

INSERT INTO registeredusers (emp_id, username, userpassword, firstname, lastname, email, created_timestamp,
is_active)
VALUES
	('ID10001', 'jdoe', 'doe10001', 'Jane', 'Doe', 'jane.doe@mail.com', NOW(), 1),
	('ID12345', 'hroberts', 'roberts12345', 'Henry', 'Roberts', 'henry.roberts@mail.com', NOW(), 1),
	('ID88888', 'gpayne', 'payne88888', 'Grace', 'Payne', 'grace.payne@mail.com', NOW(), 1),
	('ID54321', 'sjones', 'jones54321', 'Sophia', 'Jones', 'sophia.jones@mail.com', NOW(), 1),
	('ID87000', 'ssmith', 'ssmith87000', 'Sam', 'Smith', 'sam.smith@mail.com', NOW(), 0);

SELECT * FROM registeredusers;

CREATE TABLE IF NOT EXISTS public.shops
(
	outletid NUMERIC(16) PRIMARY KEY,
	ad_longitude DECIMAL(7,4) NOT NULL,
		CHECK (ad_longitude BETWEEN 103.6000 AND 104.0500), --Singapore's long bounding box
	ad_latitude DECIMAL(5,4) NOT NULL,
		CHECK (ad_latitude BETWEEN 1.2200 AND 1.4700), --Singapore's lat bounding box
	shopsize TEXT NOT NULL,
		CHECK (shopsize IN('XSMALL', 'SMALL', 'MEDIUM', 'LARGE', 'XLARGE')),
	retailer TEXT NOT NULL,
		CHECK (retailer IN('TANGS', 'BHG', 'COURTS', 'HARVEY NORMAN', 'GAIN CITY', 'BEST
DENKI', 'MUSTAFA', 'PARISILK ELECTRONICS', 'AUDIO HOUSE', 'MEGA DISCOUNT', 'MAYER
MARKETING')),
	channel VARCHAR(64) NOT NULL,
		CHECK (channel IN('Department Stores', 'Electrical Specialists', 'Technical Superstores')),
	region TEXT NOT NULL,
```

```
                    CHECK (region IN('CENTRAL REG SG', 'EAST REGION SG', 'NORTH EAST SG', 'NORTH REG SG', 'WEST REG SG')),
          copies TEXT NOT NULL,
                    CHECK (copies IN('REGULAR', 'MODELED', 'COPY_AND_CREATE')),
          ext_factor DECIMAL(4, 2) NOT NULL,
          audit_date DATE DEFAULT NOW(),
          is_active INT DEFAULT 1,
                    CHECK (is_active IN(0,1))
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.shops
   OWNER to postgres;

INSERT INTO shops (outletid, ad_longitude, ad_latitude, shopsize, retailer, channel, region, copies, ext_factor, audit_date, is_active)
VALUES
          (1427, 103.802352, 1.3412, 'XLARGE', 'TANGS', 'Department Stores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2025-01-07', 1),
          (1867, 103.845223, 1.311467, 'XLARGE', 'BHG', 'Department Stores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2025-04-21', 1),
          (4884, 103.833304, 1.298835, 'LARGE', 'COURTS', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2025-01-04', 1),
          (4601, 103.806303, 1.315835, 'LARGE', 'HARVEY NORMAN', 'Electrical Specialists', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2024-12-14', 1),
          (2795, 103.805664, 1.336388, 'LARGE', 'GAIN CITY', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2025-04-21', 1),
          (3060, 103.803095, 1.35077, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2024-10-09', 1),
          (4885, 103.8102, 1.345105, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2024-10-11', 1),
          (5155, 103.867832, 1.286522, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2024-11-23', 1),
          (3047, 103.840389, 1.351732, 'LARGE', 'COURTS', 'Technical Superstores', 'CENTRAL REG SG', 'REGULAR', 1, '2024-10-09', 1),
          (3064, 103.843208, 1.308872, 'LARGE', 'BEST DENKI', 'Technical Superstores', 'CENTRAL REG SG', 'REGULAR', 1, '2025-04-18', 1),
          (5111, 103.806338, 1.304051, 'XLARGE', 'HARVEY NORMAN', 'Technical Superstores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2025-04-30', 1),
          (4596, 103.815131, 1.306433, 'XLARGE', 'HARVEY NORMAN', 'Technical Superstores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2024-11-02', 1),
          (1423, 103.855228, 1.31406, 'XLARGE', 'MUSTAFA', 'Department Stores', 'CENTRAL REG SG', 'REGULAR', 1, '2025-01-06', 1),
          (1554, 103.841048, 1.323148, 'LARGE', 'COURTS', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2025-02-13', 1),
          (3056, 103.849038, 1.33776, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2024-11-23', 1),
```

(3063, 103.84157, 1.355218, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'CENTRAL REG SG', 'REGULAR', 1, '2024-11-23', 1),
(4000, 103.812795, 1.346215, 'LARGE', 'PARISILK ELECTRONICS', 'Electrical Specialists', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2024-11-24', 1),
(4001, 103.806444, 1.280512, 'LARGE', 'PARISILK ELECTRONICS', 'Electrical Specialists', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2025-04-30', 1),
(0871, 103.835949, 1.339235, 'LARGE', 'AUDIO HOUSE', 'Technical Superstores', 'CENTRAL REG SG', 'REGULAR', 1, '2025-01-11', 1),
(4693, 103.819609, 1.29257, 'XLARGE', 'HARVEY NORMAN', 'Technical Superstores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2025-03-29', 1),
(3823, 103.824982, 1.32759, 'XLARGE', 'MEGA DISCOUNT', 'Technical Superstores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2024-11-02', 1),
(2860, 103.833245, 1.337374, 'LARGE', 'COURTS', 'Technical Superstores', 'CENTRAL REG SG', 'REGULAR', 1, '2025-04-25', 1),
(3065, 103.860128, 1.323565, 'LARGE', 'BEST DENKI', 'Technical Superstores', 'CENTRAL REG SG', 'REGULAR', 1, '2024-12-14', 1),
(3723, 103.842953, 1.29965, 'LARGE', 'MEGA DISCOUNT', 'Technical Superstores', 'CENTRAL REG SG', 'COPY_AND_CREATE', 1, '2024-11-02', 1),
(1526, 103.938311, 1.328547, 'LARGE', 'COURTS', 'Electrical Specialists', 'EAST REGION SG', 'REGULAR', 1, '2025-04-18', 1),
(4834, 103.979173, 1.35967, 'LARGE', 'COURTS', 'Electrical Specialists', 'EAST REGION SG', 'REGULAR', 1, '2024-11-24', 1),
(3727, 103.946574, 1.336707, 'LARGE', 'HARVEY NORMAN', 'Electrical Specialists', 'EAST REGION SG', 'COPY_AND_CREATE', 1, '2024-11-30', 1),
(4629, 103.940629, 1.356958, 'LARGE', 'GAIN CITY', 'Electrical Specialists', 'EAST REGION SG', 'REGULAR', 1, '2025-03-04', 1),
(3931, 103.984461, 1.358403, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'EAST REGION SG', 'REGULAR', 1, '2025-03-04', 1),
(1034, 103.954404, 1.31002, 'LARGE', 'PARISILK ELECTRONICS', 'Electrical Specialists', 'EAST REGION SG', 'COPY_AND_CREATE', 1, '2024-10-11', 1),
(2862, 103.96688, 1.313009, 'LARGE', 'COURTS', 'Technical Superstores', 'EAST REGION SG', 'REGULAR', 1, '2024-10-05', 1),
(4597, 103.982303, 1.319521, 'XLARGE', 'HARVEY NORMAN', 'Technical Superstores', 'EAST REGION SG', 'COPY_AND_CREATE', 1, '2024-11-30', 1),
(3124, 103.930695, 1.401124, 'LARGE', 'COURTS', 'Electrical Specialists', 'NORTH EAST SG', 'REGULAR', 1, '2024-11-30', 1),
(5110, 103.890177, 1.373241, 'LARGE', 'HARVEY NORMAN', 'Electrical Specialists', 'NORTH EAST SG', 'COPY_AND_CREATE', 2, '2025-01-06', 1),
(3729, 103.920405, 1.401004, 'LARGE', 'HARVEY NORMAN', 'Electrical Specialists', 'NORTH EAST SG', 'COPY_AND_CREATE', 1, '2025-01-11', 1),
(4305, 103.913487, 1.366875, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'NORTH EAST SG', 'REGULAR', 1, '2025-01-02', 1),
(2797, 103.924826, 1.37195, 'XLARGE', 'GAIN CITY', 'Electrical Specialists', 'NORTH EAST SG', 'REGULAR', 1, '2025-04-03', 1),
(2861, 103.915677, 1.380607, 'LARGE', 'COURTS', 'Technical Superstores', 'NORTH EAST SG', 'REGULAR', 1, '2024-11-27', 1),
(1500, 103.832374, 1.427815, 'LARGE', 'COURTS', 'Electrical Specialists', 'NORTH REG SG', 'REGULAR', 1, '2025-04-18', 1),

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	(0725, 103.808342, 1.441565, 'LARGE', 'PARISILK ELECTRONICS', 'Electrical Specialists', 'NORTH REG SG', 'COPY_AND_CREATE', 1, '2024-10-09', 1),
	(4223, 103.783166, 1.452982, 'XLARGE', 'GAIN CITY', 'Technical Superstores', 'NORTH REG SG', 'REGULAR', 1, '2025-01-06', 1),
	(1499, 103.706356, 1.312101, 'LARGE', 'COURTS', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2024-10-09', 1),
	(3879, 103.697198, 1.325327, 'LARGE', 'COURTS', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2025-03-21', 1),
	(3454, 103.769089, 1.322662, 'LARGE', 'COURTS', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2024-10-09', 1),
	(2798, 103.737935, 1.33272, 'LARGE', 'COURTS', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2024-11-02', 1),
	(3730, 103.746592, 1.34298, 'LARGE', 'HARVEY NORMAN', 'Electrical Specialists', 'WEST REG SG', 'COPY_AND_CREATE', 1, '2024-11-15', 1),
	(3059, 103.749874, 1.383734, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2025-03-21', 1),
	(4881, 103.703968, 1.389436, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2024-10-09', 1),
	(4559, 103.743771, 1.344337, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2025-03-04', 1),
	(3262, 103.736273, 1.372632, 'LARGE', 'BEST DENKI', 'Electrical Specialists', 'WEST REG SG', 'REGULAR', 1, '2025-01-11', 1),
	(3771, 103.699656, 1.370819, 'MEDIUM', 'MAYER MARKETING', 'Electrical Specialists', 'WEST REG SG', 'COPY_AND_CREATE', 1, '2024-12-27', 1);

SELECT * FROM shops;

CREATE TABLE IF NOT EXISTS public.retailers
(
	retailer VARCHAR(64) PRIMARY KEY,
		CHECK (retailer IN('TANGS', 'BHG', 'COURTS', 'HARVEY NORMAN', 'GAIN CITY', 'BEST DENKI', 'MUSTAFA', 'PARISILK ELECTRONICS', 'AUDIO HOUSE', 'MEGA DISCOUNT', 'MAYER MARKETING')),
	orgtype VARCHAR(16) NOT NULL,
		CHECK (orgtype IN('CHAINS', 'INDEPENDENTS'))
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.retailers
   OWNER to postgres;

INSERT INTO retailers (retailer, orgtype)
VALUES
	('MUSTAFA', 'INDEPENDENTS'),
	('TANGS', 'INDEPENDENTS'),
	('BHG', 'INDEPENDENTS'),
	('COURTS', 'CHAINS'),
	('BEST DENKI', 'CHAINS'),
```

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	('PARISILK ELECTRONICS', 'CHAINS'),
	('HARVEY NORMAN', 'CHAINS'),
	('GAIN CITY', 'CHAINS'),
	('MAYER MARKETING', 'INDEPENDENTS'),
	('AUDIO HOUSE', 'CHAINS'),
	('MEGA DISCOUNT', 'CHAINS');

SELECT * FROM retailers;


CREATE TABLE IF NOT EXISTS public.extrapolationcell
(
	shopsize TEXT NOT NULL,
		CHECK (shopsize IN('XSMALL', 'SMALL', 'MEDIUM', 'LARGE', 'XLARGE')),
	orgtype VARCHAR(16) NOT NULL,
		CHECK (orgtype IN('CHAINS', 'INDEPENDENTS')),
	channel VARCHAR(64) NOT NULL,
		CHECK (channel IN('Department Stores', 'Electrical Specialists', 'Technical Superstores')),
	region TEXT NOT NULL,
		CHECK (region IN('CENTRAL REG SG', 'EAST REGION SG', 'NORTH EAST SG', 'NORTH
REG SG', 'WEST REG SG')),
	ext_factor DECIMAL(4, 2) NOT NULL,
	PRIMARY KEY (shopsize, orgtype, channel, region)
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.extrapolationcell
   OWNER to postgres;

INSERT INTO extrapolationcell (shopsize, orgtype, channel, region, ext_factor )
VALUES
('LARGE', 'CHAINS', 'Electrical Specialists', 'EAST REGION SG', 1.00),
('XLARGE', 'CHAINS', 'Electrical Specialists', 'NORTH EAST SG', 1.00),
('LARGE', 'CHAINS', 'Technical Superstores', 'NORTH EAST SG', 1.00),
('LARGE', 'CHAINS', 'Electrical Specialists', 'NORTH EAST SG', 1.25),
('LARGE', 'CHAINS', 'Technical Superstores', 'CENTRAL REG SG', 1.00),
('LARGE', 'CHAINS', 'Electrical Specialists', 'CENTRAL REG SG', 1.00),
('XLARGE', 'CHAINS', 'Technical Superstores', 'NORTH REG SG', 1.00),
('LARGE', 'CHAINS', 'Electrical Specialists', 'WEST REG SG', 1.00),
('XLARGE', 'CHAINS', 'Technical Superstores', 'EAST REGION SG', 1.00),
('LARGE', 'CHAINS', 'Technical Superstores', 'EAST REGION SG', 1.00),
('XLARGE', 'INDEPENDENTS', 'Department Stores', 'CENTRAL REG SG', 1.00),
('LARGE', 'CHAINS', 'Electrical Specialists', 'NORTH REG SG', 1.00),
('MEDIUM', 'INDEPENDENTS', 'Electrical Specialists', 'WEST REG SG', 1.00),
('XLARGE', 'CHAINS', 'Technical Superstores', 'CENTRAL REG SG', 1.00);


SELECT * FROM extrapolationcell;
```

```
CREATE TABLE IF NOT EXISTS public.items
(
	itemid SERIAL PRIMARY KEY,
	itemname VARCHAR(64) UNIQUE NOT NULL,
	productgroup VARCHAR(64) NOT NULL,
	brand VARCHAR(32) NOT NULL
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.items
    OWNER to postgres;

INSERT INTO items (itemid, itemname, productgroup, brand)
VALUES
	(149976736, 'DFB425FP', 'DISHWASHERS', 'LG'),
	(158043996, 'EBDW 1351 A WH', 'DISHWASHERS', 'ELBA'),
	(146597283, 'DFN05X11W', 'DISHWASHERS', 'BEKO'),
	(171641414, 'WFC3C26XAUS', 'DISHWASHERS', 'WHIRLPOOL'),
	(156338670, 'EDW 3050U', 'DISHWASHERS', 'EUROPACE'),
	(162803615, 'SMS46GW01P', 'DISHWASHERS', 'BOSCH'),
	(97823566, 'SMS63L02EA', 'DISHWASHERS', 'BOSCH'),
	(58664210, 'SMS50E82EU', 'DISHWASHERS', 'BOSCH'),
	(167766976, 'SMS2HAI12E', 'DISHWASHERS', 'BOSCH'),
	(143315315, 'SMS46IW20E', 'DISHWASHERS', 'BOSCH'),
	(165965380, 'SMS4ECI14E', 'DISHWASHERS', 'BOSCH'),
	(165618063, 'SMS8YCI01E', 'DISHWASHERS', 'BOSCH'),
	(165614392, 'SMI4HCS48E', 'DISHWASHERS', 'BOSCH'),
	(165866919, 'SMV6ZCX42E', 'DISHWASHERS', 'BOSCH'),
	(102485644, 'G 4920 SC', 'DISHWASHERS', 'MIELE'),
	(168386590, 'G 7360 C SCVI AUTODOS', 'DISHWASHERS', 'MIELE'),
	(155068774, 'G 7100 SC', 'DISHWASHERS', 'MIELE'),
	(168179252, 'G 7310 C SC AUTODOS', 'DISHWASHERS', 'MIELE'),
	(166661884, 'G 7310 C SCI AUTODOS', 'DISHWASHERS', 'MIELE'),
	(94681039, 'G 4960 SCVI', 'DISHWASHERS', 'MIELE'),
	(117971190, 'G 6993 SCVI K2O', 'DISHWASHERS', 'MIELE'),
	(166357990, 'SPS2HKW57E', 'DISHWASHERS', 'BOSCH'),
	(133569980, 'ESF6010BW', 'DISHWASHERS', 'ELECTROLUX'),
	(116300240, 'DD60SCTX9', 'DISHWASHERS', 'FISHER&PAYKEL'),
	(140199895, 'DD60STI9', 'DISHWASHERS', 'FISHER&PAYKEL'),
	(116156915, 'DD60DCX9', 'DISHWASHERS', 'FISHER&PAYKEL'),
	(116506432, 'DD60SCX9', 'DISHWASHERS', 'FISHER&PAYKEL'),
	(147284348, 'DFB325HS', 'DISHWASHERS', 'LG'),
	(147263475, 'DFB227HD', 'DISHWASHERS', 'LG'),
	(164516251, 'DWF128DW', 'DISHWASHERS', 'BRANDT'),
	(163010648, 'SKS62E32EU', 'DISHWASHERS', 'BOSCH'),
	(165827964, 'SPI2HKS59E', 'DISHWASHERS', 'BOSCH'),
	(125089977, 'VH1772J', 'DISHWASHERS', 'BRANDT'),
```

```sql
	(116120838, 'SMV68TX06E', 'DISHWASHERS', 'BOSCH'),
	(171511072, 'SMS2IVW01P', 'DISHWASHERS', 'BOSCH'),
	(159831788, 'DFB227HM', 'DISHWASHERS', 'LG'),
	(166588328, 'EBDW 1481M WH', 'DISHWASHERS', 'ELBA'),
	(156116514, 'CDPN2D360PW', 'DISHWASHERS', 'CANDY'),
	(168172066, 'CDPN 1L390PW-80', 'DISHWASHERS', 'CANDY'),
	(157615298, 'CDPN2D522PX', 'DISHWASHERS', 'CANDY'),
	(159694565, 'CDPMN 4S622PX', 'DISHWASHERS', 'CANDY'),
	(122212884, 'WFC 3C26 F UK', 'DISHWASHERS', 'WHIRLPOOL'),
	(124184210, 'ESF5206LOW', 'DISHWASHERS', 'ELECTROLUX'),
	(165189495, 'DWF137DS', 'DISHWASHERS', 'BRANDT'),
	(125317377, 'VH1772X', 'DISHWASHERS', 'BRANDT'),
	(166933759, 'G 7100 C SC', 'DISHWASHERS', 'MIELE'),
	(140199652, 'DD60SI9', 'DISHWASHERS', 'FISHER&PAYKEL'),
	(137588290, 'DD60DI9', 'DISHWASHERS', 'FISHER&PAYKEL'),
	(163079407, 'DWS-22ASG(K)', 'DISHWASHERS', 'TOSHIBA'),
	(115507221, 'SMV46MX03E', 'DISHWASHERS', 'BOSCH'),
	(93414175, 'SMS88TI03E', 'DISHWASHERS', 'BOSCH'),
	(157227747, 'CDPN 1L390PW', 'DISHWASHERS', 'CANDY'),
	(134027714, 'ESF5512LOX', 'DISHWASHERS', 'ELECTROLUX'),
	(117616374, 'SMS46MI05E', 'DISHWASHERS', 'BOSCH'),
	(165475613, 'SMU4HCS48E', 'DISHWASHERS', 'BOSCH'),
	(116067431, 'SMI46MS03E', 'DISHWASHERS', 'BOSCH'),
	(134315681, 'ESL5343LO', 'DISHWASHERS', 'ELECTROLUX'),
	(142677619, 'DVH1323J', 'DISHWASHERS', 'DE DIETRICH');

SELECT * FROM items;

CREATE TABLE IF NOT EXISTS public.sales
(
	outletid NUMERIC(16),
	itemid NUMERIC(64) NOT NULL,
	units NUMERIC(8) NOT NULL DEFAULT 0,
	price DECIMAL(8,2) NOT NULL DEFAULT 0,
	PRIMARY KEY(outletid, itemid)
)

TABLESPACE pg_default;

ALTER TABLE IF EXISTS public.sales
   OWNER to postgres;

INSERT INTO sales (outletid, itemid, units, price)
VALUES
	(725, 97823566, 3, 956.33),
	(725, 122212884, 2, 1295.8),
	(725, 124184210, 1, 859),
	(725, 133569980, 2, 759),
	(725, 134027714, 1, 1309),
```

(725, 140199652, 1, 1399),
(725, 143315315, 1, 1394),
(725, 146597283, 1, 599),
(725, 147284348, 3, 1739),
(725, 162803615, 3, 1359),
(725, 163010648, 1, 1034),
(725, 167766976, 1, 1379.01),
(871, 58664210, 2, 809),
(871, 97823566, 39, 980),
(871, 116120838, 3, 2699),
(871, 122212884, 1, 1295.8),
(871, 133569980, 29, 759),
(871, 143315315, 6, 1399),
(871, 146597283, 2, 599),
(871, 147284348, 33, 1599),
(871, 149976736, 20, 1499),
(871, 156116514, 10, 841.5),
(871, 157615298, 2, 1106.7),
(871, 158043996, 1, 542),
(871, 159694565, 2, 1440),
(871, 159831788, 10, 1919),
(871, 162803615, 4, 1359),
(871, 163010648, 3, 978.99),
(871, 165614392, 4, 2090),
(871, 165866919, 1, 2829.01),
(871, 165965380, 3, 2239),
(871, 166357990, 1, 1549),
(871, 166588328, 21, 629),
(871, 167766976, 5, 1199),
(871, 168172066, 23, 652.8),
(871, 171511072, 1, 1219),
(1034, 133569980, 1, 759),
(1034, 162803615, 2, 1359),
(1423, 146597283, 6, 535),
(1423, 149976736, 7, 1449),
(1423, 156338670, 1, 459),
(1423, 158043996, 1, 559),
(1423, 171641414, 1, 949),
(1427, 58664210, 12, 809),
(1427, 94681039, 1, 2000),
(1427, 97823566, 46, 852.8),
(1427, 102485644, 5, 1039),
(1427, 117971190, 1, 4700),
(1427, 143315315, 2, 1399),
(1427, 155068774, 9, 2299),
(1427, 162803615, 19, 1255),
(1427, 165618063, 2, 2832),
(1427, 165866919, 2, 2829.01),
(1427, 165965380, 4, 2239),

(1427, 166661884, 5, 3051),
(1427, 167766976, 6, 1379.01),
(1427, 168179252, 4, 2799),
(1427, 168386590, 9, 3431.83),
(1427, 165614392, 2, 2090),
(1499, 97823566, 2, 924),
(1499, 162803615, 2, 1309),
(1499, 163079407, 2, 459),
(1500, 97823566, 7, 956.33),
(1500, 122212884, 3, 1295.8),
(1500, 124184210, 1, 859),
(1500, 133569980, 1, 759),
(1500, 134027714, 1, 1309),
(1500, 140199652, 1, 1399),
(1500, 143315315, 1, 1394),
(1500, 146597283, 1, 599),
(1500, 147284348, 4, 1739),
(1500, 162803615, 7, 1359),
(1500, 163010648, 1, 1034),
(1500, 167766976, 1, 1379.01),
(1526, 133569980, 1, 759),
(1526, 162803615, 2, 1359),
(1554, 58664210, 1, 899),
(1554, 97823566, 5, 847),
(1554, 116156915, 3, 2301.5),
(1554, 116300240, 1, 1680),
(1554, 116506432, 1, 1449),
(1554, 133569980, 2, 739),
(1554, 140199895, 1, 1499),
(1554, 149976736, 5, 1375),
(1554, 162803615, 5, 1359),
(1554, 166357990, 1, 1699),
(1554, 167766976, 1, 1459),
(1867, 146597283, 4, 535),
(1867, 149976736, 3, 1449),
(1867, 156338670, 1, 459),
(1867, 158043996, 1, 559),
(1867, 171641414, 1, 949),
(2795, 58664210, 6, 809),
(2795, 97823566, 16, 852.8),
(2795, 102485644, 2, 1039),
(2795, 124184210, 1, 779),
(2795, 133569980, 3, 689.01),
(2795, 158043996, 4, 542),
(2795, 159831788, 2, 1919),
(2795, 162803615, 3, 1151),
(2795, 163079407, 1, 459),
(2795, 164516251, 1, 609),
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